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2027 CYCLE

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CUET UG

General Aptitude Test

Numerical Ability & Data Interpretation

Official Syllabus · General Test (Section III) · Class 12 · CUET UG 2027 Cycle Notes

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Syllabus Alignment

Sub-topic	Official Syllabus Pointer (NTA CUET GAT 501)	Core Skills Tested
Percentages & Variation	Percentages and quantitative changes: Fractions, successive shifts, profit/loss, reverse base.	Rapid fraction conversions, base identification, successive variation formula ($a + b + a\frac{b}{100}$).
Ratios & Partnerships	Ratios, tildeortions, and partnerships: Part allocation, variations, capital-time compounding.	Splitting quantities, tildeortional scaling, investment duration weighting, alligation cross.
Averages & Mixtures	Averages and central tendencies: Totals manipulation, joining/leaving replacements, weighted means.	Translating averages to aggregate sums, deviation shortcuts, weighted arithmetic means.
Data Interpretation	Simple graphs/tables/charts reading: Bar graphs, line charts, pie charts, cross-tabulation.	Rapid frame triage, angle-to-percent conversion ($3.6° = 1\%$), tabular coordinate extraction.
Basic Statistics	Basic statistics interpretation: Mean, median, mode, range, and outlier sensitivity.	Sorted median determination, empirical modal relation, identifying skewed distributions.

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EXAM PATTERN

CUET UG 2027 Official Exam Pattern: The General Test (Section III, Code 501) presents **50 compulsory multiple-choice questions** to be attempted in **60 minutes**. Each correct response yields +5 marks, while an incorrect response incurs a −1 mark negative deduction (unattempted questions earn 0). Numerical Ability and Data Interpretation constitutes approximately **15 to 20 questions** of the test. The numerical problems are engineered to reward mental arithmetic, fraction anchors, and structural estimation rather than tedious long division.

Percentages: The Foundational Multiplier

A percentage is mathematically equivalent to a rational fraction scaled to a denominator of 100. Competitive examination candidates who execute percentage calculations through multi-digit decimal multiplication squander valuable time. Mastering standard fraction anchors reduces complex percentage operations to instantaneous single-step divisions.

Standard Fraction-to-Percentage Equivalence Table

Fraction	1/2	1/3	1/4	1/5	1/6	1/8	1/12
Percent	50.0%	33.33%	25.0%	20.0%	16.67%	12.5%	8.33%

Mental Arithmetic Levers: The 10% Build-Up Method

Any composite percentage can be rapidly synthesized by decomposing it into standard base units:

- **10% Base:** Divide the target integer by 10 (shift the decimal point one place to the left).
- **5% Base:** Halve the 10% value.
- **1% Base:** Divide the target integer by 100 (shift the decimal point two places to the left).
- **Synthesis Example:** To evaluate 17.5% of 480:

$$10\% = 48, \quad 5\% = 24, \quad 2.5\% = 12 \Longrightarrow 17.5\% = 48 + 24 + 12 = 84.$$

The Percentage Change Base Rule

A recurrent error in data interpretation stems from misidentifying the denominator in percentage change queries. The formula strictly demands calibration against the **original (initial) reference value**:

$$\text{Percentage Change} = \frac{\text{Final Value} - \text{Initial Value}}{\text{Initial Value}} \times 100\%$$

- If revenue expands from ₹40 Lakhs to ₹50 Lakhs:

$$\%\ \text{Increase} = \frac{50 - 40}{40} \times 100\% = \frac{10}{40} \times 100\% = 25.0\%.$$

(Dividing by the terminal value 50 yields an erroneous 20.0%).

Successive Percentage Variations

When a quantity undergoes two consecutive percentage alterations of $a\%$ and $b\%$, the net compound percentage variation is given by:

$$\text{Net Percentage Variation} = \left(a + b + \frac{a \cdot b}{100}\right)\%$$

- **The Classic Neutralization Myth:** A price hike of $+20\%$ followed by a discount of -20% does NOT restore the original price.

$$\text{Net Variation} = (+20) + (-20) + \frac{(+20)(-20)}{100} = 0 - \frac{400}{100} = -4.0\%.$$

The quantity suffers a net reduction of 4.0% because the second percentage deduction operates upon a larger intermediate base.

Reverse Percentage Arithmetic

When given a terminal value after an $x\%$ appreciation or depreciation:

- If a commodity sells for ₹15,000 following a 25% markup:

$$\text{Selling Price} = \text{Cost Price} \times \left(1 + \frac{25}{100}\right) = \text{Cost Price} \times \frac{5}{4}$$

$$\text{Cost Price} = 15,000 \times \frac{4}{5} = 12,000 \ \text{ rupees.}$$

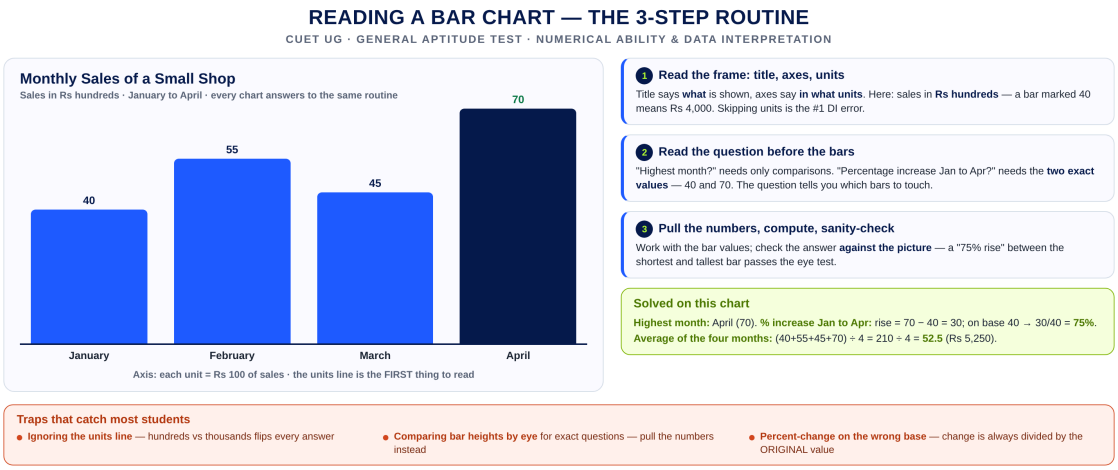


Figure 1: Systematic extraction workflow for Data Interpretation: Highlighting frame triage, coordinate mapping, fraction conversions, and the eye-check validation routine.

Ratios, Proportions, and Commercial Partnerships

A ratio represents the quantitative comparison of two or more homogeneous quantities expressed in minimal integral form.

Partitioning Quantities into Ratios

To distribute a total corpus T among recipients in the ratio $a : b : c$:

1. Calculate the aggregate sum of parts: $S = a + b + c$.
2. Determine the unitary value of a single part: $U = \frac{T}{S}$.
3. Compute individual allocations:
$$\text{Share of } A = a \cdot U, \quad \text{Share of } B = b \cdot U, \quad \text{Share of } C = c \cdot U.$$
4. **Internal Verification:** The sum of individual shares must perfectly reconstruct the original total T.

Commercial Partnership Equations

In business partnerships, profit distribution is tildeortional to both the invested capital (C) and the operational time duration (T):

$$\text{Profit Ratio} = (C_A \cdot T_A) : (C_B \cdot T_B) : (C_C \cdot T_C)$$

- If Partner A invests ₹50,000 for 12 months and Partner B invests ₹75,000 for 8 months:
$$\text{Ratio} = (50,000 \times 12) : (75,000 \times 8) = 6,00,000 : 6,00,000 = 1 : 1.$$
Both partners receive identical profit shares despite differing capital inputs.

The Alligation Rule for Mixtures

When two ingredients of unit costs C (cheaper) and D (dearer) are combined to produce a mixture of intermediate mean cost M:

$$\frac{Q_{\text{Cheaper}}}{Q_{\text{Dearer}}} = \frac{D - M}{M - C}$$

ALLIGATION IN ACTION

Problem: In what ratio must grocer blend Darjeeling tea worth ₹280/kg with Assam tea worth ₹350/kg to create a blend worth ₹300/kg?

- Cheaper price $C = 280$, Dearer price $D = 350$, Mean price $M = 300$.
- Difference on Dearer side: $D - M = 350 - 300 = 50$.
- Difference on Cheaper side: $M - C = 300 - 280 = 20$.

$$\text{Ratio} = 50 : 20 = 5 : 2.$$

The grocer must mix 5 parts of Darjeeling tea with 2 parts of Assam tea.

Averages and Weighted Combinations

An arithmetic average represents the balanced equilibrium point of a numeric distribution:

$$\text{Average } (A) = \frac{\text{Sum of Observations } (\Sigma x)}{\text{Total Count } (N)} \implies \Sigma x = A \cdot N$$

Almost all competitive exam puzzles resolve cleanly the moment averages are converted into **aggregate sums**.

The Replacement / Addition Dynamic

When an element of value V_{new} replaces an existing element V_{old}:

$$\text{New Sum} = \text{Old Sum} - V_{\text{old}} + V_{\text{new}}$$

$$\text{New Average} = \text{Old Average} + \frac{V_{\text{new}} - V_{\text{old}}}{N}$$

- If a player weighing 85 kg replaces a player weighing 65 kg in a team of 10:

$$\Delta A = \frac{85 - 65}{10} = +2.0 \text{ kg rise in team average.}$$

Weighted Average Formulation

When two distinct sub-groups of sizes n_1 and n_2 possessing independent group averages A_1 and A_2 are merged:

$$A_{\text{weighted}} = \frac{n_1 A_1 + n_2 A_2}{n_1 + n_2}$$

- **Pitfall:** One cannot simply average the averages $(A_1 + \frac{A_2}{2})$ unless both sub-groups contain an identical number of observations $(n_1 = n_2)$.

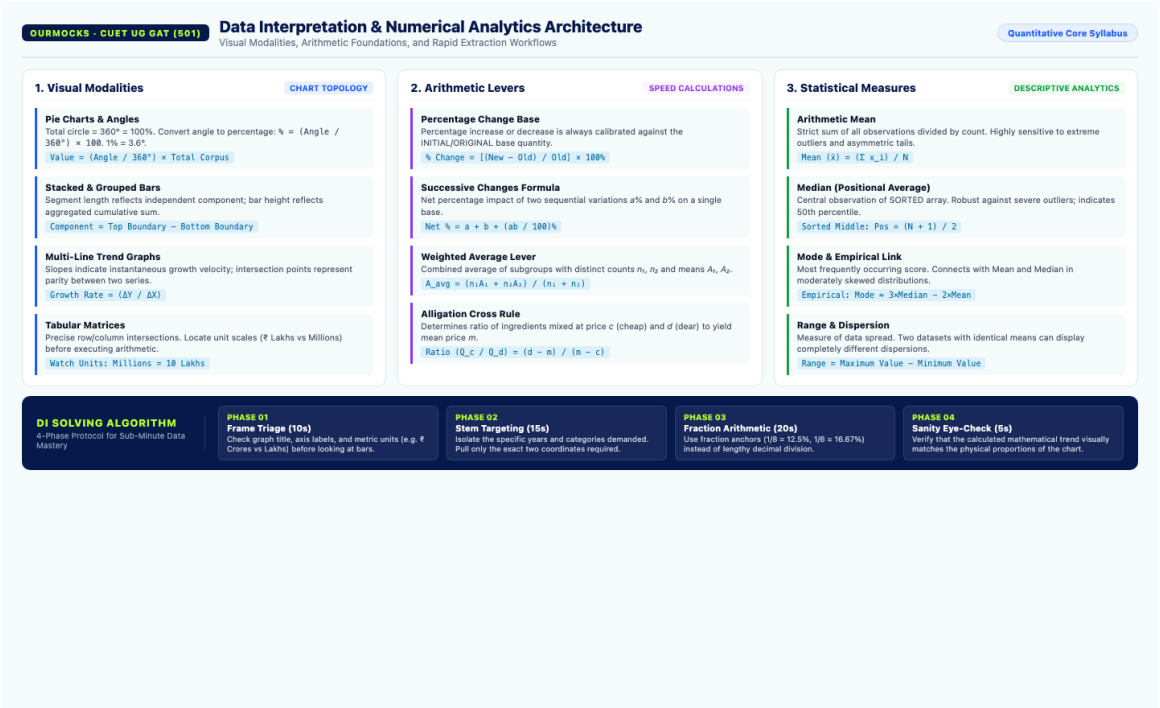


Figure 2: Comprehensive architectural taxonomy of Data Interpretation: Illustrating chart modalities, arithmetic multipliers, statistical measures, and the 4-phase solving algorithm.

Modalities of Data Interpretation

CUET questions present empirical data across five standard visual modalities:

1. Pie Charts (Circular Proportions)

Pie charts portray the allocation of a single holistic corpus divided into interors. Sectors are denominated in either percentages or central angles:

- **Angular Conversion Metric:** A complete circular disc subtends 360°, corresponding to 100%.

$$100\% = 360^\circ \implies 1\% = 3.6^\circ, \qquad 1^\circ = \frac{100}{360}\% = \frac{5}{18}\% \approx 0.2778\%.$$

  - Sector of $54^\circ = \frac{54}{3.6}\% = 15.0\%$.
  - Sector of $72^\circ = \frac{72}{3.6}\% = 20.0\%$.
  - Sector of $90^\circ = \frac{90}{3.6}\% = 25.0\%$.
- **Double Pie Charts:** When two pie charts represent different calendar years, interor sizes **cannot** be directly compared without factoring in the total baseline expenditure of each respective year.

2. Bar Charts (Categorical Comparisons)

- **Simple Vertical/Horizontal Bars:** Best suited for comparing discrete independent categories (e.g., annual revenue across five corporate divisions).
- **Grouped Bars:** Displays multi-attribute data side-by-side for each category (e.g., Exports vs Imports over consecutive financial years).
- **Stacked (Sub-divided) Bars:** The entire bar represents total volume, while internal color segments depict component contributions. To extract a component value, compute the difference between the segment's upper and lower boundaries:

$$\text{Component Value} = \text{Upper Boundary Coordinate} - \text{Lower Boundary Coordinate}.$$

3. Line Graphs (Continuous Time-Series)

Line graphs illustrate continuous dynamic fluctuations over temporal intervals (months, quarters, years):

- **Rate of Growth (Slope):** The steepness of the line segment indicates acceleration or deceleration.
- **Crossover Interinterions:** The exact coordinate where two line curves interinter signifies parity between the two tracked entities.

4. Tabular Cross-Tabulation Matrices

Tables provide exact numerical values across rows and columns without optical estimation:

- Always scan the row headers and column banners before extracting data.
- Check unit annotations carefully: confusing **Metric Tonnes** with **Kilograms** or **₹ Crores** with **₹ Lakhs** (1 Crore = 100 Lakhs) is a classic distractor trap.

Basic Statistics and Data Dispersion

The official NTA syllabus tests foundational descriptive statistical parameters and their structural interactions.

Metric	Mathematical Definition	Operational Characteristics & Traps
Arithmetic Mean	Sum of all data points divided by sample size: $\bar{x} = \frac{\Sigma x_i}{N}$	Default central measure. Highly sensitive to extreme outlier values; distorted in skewed distributions.
Median	Middle value of array after STRICT SORTING in ascending order.	Positional average. Completely resistant to outlier extremes. Identifies the 50th percentile rank.
Mode	The observation appearing with the highest absolute frequency.	Can be unimodal, bimodal, or non-existent. Most practical metric for commercial inventory planning.
Range	Difference between extremes: Range $= X_{\max} - X_{\min}$	Simplest measure of statistical dispersion. Sensitive only to the extreme boundaries.

The Median Sorting Mandate

To evaluate the median of an array:

1. Sort all N observations in non-decreasing order.
2. If N is odd, the median occupies position $\frac{N+1}{2}$.
3. If N is even, the median is the arithmetic mean of observations at positions $\frac{N}{2}$ and $\left(\frac{N}{2}\right)+1$.

- **Warning:** Selecting the middle number of an unsorted list is the most common distractor in CUET statistics.

The Empirical Pearsonian Relationship

For moderately skewed, unimodal distributions, an approximate empirical balance links the three central measures:

$$\text{Mode} \approx 3 \times \text{Median} - 2 \times \text{Mean}$$

- If a dataset possesses a Median of 25.0 and a Mean of 22.0:

$$\text{Estimated Mode} \approx 3(25.0) - 2(22.0) = 75.0 - 44.0 = 31.0.$$

Tactical DI Speed Toolkit & Approximation Hacks

To conquer 50 questions in 60 minutes, candidates must execute calculations through disciplined structural approximation:

- **The 1% and 10% Calibration:** Avoid multi-digit denominator division. If required to calculate $\frac{143}{682}$:
  - 10% of $682 = 68.2$.
  - 20% of $682 = 136.4$.
  - Difference $= 143 - 136.4 = 6.6$.
  - Since 1% of $682 \approx 6.8$, the ratio is approximately $20\% + 1\% = 21.0\%$.
  - This calculation takes 6 seconds mentally without scrap paper!
- **Unit Scaling Vigilance:** In NTA papers, graphs often print labels like **Production in '000 Units** or **Values in ₹ Millions**. An answer of 45 on the axis actually denotes $45,000$ or ₹$4,50,00,000$.

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Advanced Commercial Arithmetic: Discounts, Interest & Faulty Weights

Beyond elementary percentage variations, CUET GAT evaluates commercial arithmetic models that combine markup, discounts, and compound variations:

1. Markup, Discount, and True Profit Mechanics

Retail transactions connect Cost Price (CP), Marked Price (MP), and Selling Price (SP):

$$\text{Marked Price} = \text{CP} \times \left(1 + \frac{M}{100}\right), \qquad \text{Selling Price} = \text{MP} \times \left(1 - \frac{D}{100}\right)$$

$$\text{Overall Profit Percentage} = \frac{\text{SP} - \text{CP}}{\text{CP}} \times 100\% = \left[M - D - \frac{M \cdot D}{100}\right]\%$$

- **Successive Discounts:** If successive trade discounts of $d_1\%, d_2\%, d_3\%$ are granted:

$$\text{Net Selling Price} = \text{MP} \times \left(1 - \frac{d_1}{100}\right) \times \left(1 - \frac{d_2}{100}\right) \times \left(1 - \frac{d_3}{100}\right)$$

  - Successive discounts of 20% and 10% yield:

$$\text{Net Discount} = 20 + 10 - \frac{20 \times 10}{100} = 30 - 2 = 28.0\% \quad \text{(never 30\%!)}.$$

2. False Weights and Dishonest Dealer Problems

When a merchant claims to sell commodities at cost price but utilizes a defective weight:

$$\text{Gain Percentage} = \frac{\text{Error in Weight}}{\text{True Value} - \text{Error}} \times 100\% = \frac{\text{Claimed Weight} - \text{Actual Weight}}{\text{Actual Weight}} \times 100\%$$

- If a shopkeeper uses an 800 g weight instead of a standard 1 kg (1,000 g) measure:

$$\%\ \text{Profit} = \frac{1,000 - 800}{800} \times 100\% = \frac{200}{800} \times 100\% = 25.0\%.$$

3. Simple vs Compound Interest Baselines in GAT

- **Simple Interest (SI):** Principal P remains constant throughout tenure T:

$$\text{SI} = \frac{P \cdot R \cdot T}{100}, \qquad \text{Amount} = P + \text{SI} = P\left(1 + \frac{RT}{100}\right)$$

- **Compound Interest (CI):** Interest accrues on accumulated balance:

$$\text{CI Amount} = P\left(1 + \frac{R}{100}\right)^T, \qquad \text{CI} = \text{Amount} - P$$

- **The 2-Year and 3-Year Differential Shortcuts:**

$$\text{Difference } (\text{CI}_2 - \text{SI}_2) = P\left(\frac{R}{100}\right)^2$$

$$\text{Difference } (\text{CI}_3 - \text{SI}_3) = P\left(\frac{R}{100}\right)^2 \left[3 + \frac{R}{100}\right]$$

These shortcuts eliminate multi-line compounding calculations in competitive exams.

Proportional Variations, Work, and Rates

1. Mathematical Variations

- **Direct Proportionality:** $y \sim x \implies y = kx$. As x scales by factor λ, y scales by λ.
- **Inverse Proportionality:** $y \sim \frac{1}{x} \implies y \cdot x = k$. If x doubles, y is halved.
- **Joint Proportionality:** $W \sim (M \cdot D \cdot H) \implies \frac{M_1 D_1 H_1}{W_1} = \frac{M_2 D_2 H_2}{W_2}$ (where $M = \text{Men}, D = \text{Days}, H = \text{Hours/Day}, W = \text{Work Output}$).

2. Work, Time, and Operational Efficiency

Work output operates on the principle of individual operational efficiency:

$$\text{Efficiency } (E) = \frac{\text{Total Work}}{\text{Time Taken}} \implies \text{Time Taken} = \frac{\text{Total Work}}{E}$$

- **Combined Work Formula:** If Agent A completes work in T_A days and Agent B in T_B days:

$$\frac{1}{T_{\text{combined}}} = \frac{1}{T_A} + \frac{1}{T_B} \implies T_{\text{combined}} = \frac{T_A \cdot T_B}{T_A + T_B}$$

- **Pipes and Cisterns:** Inlet pipes perform positive work ($+\frac{1}{T_{\text{in}}}$), while drainage leaks perform negative work ($-\frac{1}{T_{\text{out}}}$):

$$\text{Net Inflow Rate} = \frac{1}{T_{\text{inlet}}} - \frac{1}{T_{\text{outlet}}}$$

Advanced Data Interpretation Modalities

1. Radar (Spider) Charts

Radar charts plot multivariate categorical data along radiating equi-angular axes extending from a central origin:

- Values furthest from the center represent maximum performance.
- The enclosed polygon area illustrates comprehensive relative capability across all evaluated parameters.

2. Caselet DI (Descriptive Narrative Conversion)

Caselet DI presents unstructured numerical data inside a dense narrative paragraph. The candidate must reconstruct a structured two-dimensional table before attempting question stems:

- **Step 1:** Identify the primary entities (rows) and attributes (columns).
- **Step 2:** Populate the matrix with explicit values directly stated in the text.
- **Step 3:** Use algebraic constraints (row totals, column sums, percentages) to solve for missing cells sequentially.

Speed, Distance, and Time Fundamentals in GAT

Time, Speed, and Distance questions frequently appear in CUET General Test sections. The governing relation connects distance (D), velocity (S), and transit time (T):

$$D = S \cdot T \implies S = \frac{D}{T} \implies T = \frac{D}{S}$$

1. Unit Conversion Metrics

- To convert kilometers per hour (km/h) to meters per second (m/s):

$$1 \text{ km/h} = 1,000 \frac{\text{m}}{3,600 \text{ s}} = \frac{5}{18} \text{ m/s}$$

  - $72 \text{ km/h} = 72 \times \frac{5}{18} = 20 \text{ m/s}$.
  - $54 \text{ km/h} = 54 \times \frac{5}{18} = 15 \text{ m/s}$.
- To convert m/s to km/h, multiply by $\frac{18}{5}$.

2. Relative Speed Principles

- **Opposite Directions (Converging / Diverging):** Relative velocity is the arithmetic sum of individual velocities:

$$S_{\text{rel}} = S_1 + S_2$$

- **Same Direction (Overtaking):** Relative velocity is the absolute positive difference between individual velocities:

$$S_{\text{rel}} = |S_1 - S_2|$$

3. Motion of Trains

- **Train Crossing a Stationary Point Object (Pole, Tree, Person):**

$$\text{Time} = \frac{\text{Length of Train } (L_T)}{\text{Speed of Train } (S_T)}$$

- **Train Crossing an Extended Object (Platform, Bridge, Tunnel):**

$$\text{Time} = \frac{L_T + L_{\text{Platform}}}{S_T}$$

- **Two Trains Crossing Each Other:**

$$\text{Time} = \frac{L_1 + L_2}{S_1 \pm S_2} \quad (+ \text{ for opposite, - for same direction}).$$

4. Boats and Streams (Hydrodynamic Motion)

Let the speed of a boat in still water be v and the speed of the water current be s:

- **Downstream Velocity (With Stream):** $v_d = v + s$.
- **Upstream Velocity (Against Stream):** $v_u = v - s$.
- **Reversing Relationships:**

$$v = \frac{v_d + v_u}{2}, \qquad s = \frac{v_d - v_u}{2}$$

- If a boat travels downstream at 18 km/h and upstream at 10 km/h:

$$v = \frac{18 + 10}{2} = 14 \text{ km/h (still water)}, \qquad s = \frac{18 - 10}{2} = 4 \text{ km/h (stream)}.$$

Chapter Synthesis & Cognitive Mindmap



Figure 3: Comprehensive architectural mindmap of Numerical Ability & Data Interpretation: Synthesizing percentage anchors, ratio partitioning, average manipulation, chart topologies, and speed analytics.

HIGH-YIELD MEMORY HOOKS FOR CUET 2027 GAT

Keep these fundamental mathematical and data interpretation maxims primed:

- **Base Discipline:** In percentage change, divide by the **original** value, never the terminal value!
- **Successive Shifts:** A hike of $x\%$ followed by a drop of $x\%$ yields an inevitable net loss of $\left(\frac{x^2}{100}\right)\%$.
- **Angle-to-Percent:** Divide degrees by 3.6 to obtain percent; multiply percent by 3.6 to obtain degrees.
- **Averages to Totals:** Never average averages! Immediately convert every average into its aggregate total ($\Sigma x = A \cdot N$).
- **Sorted Median:** Always sort the data array before identifying the central median element.
- **Partnership Capital-Time:** Profit splits strictly in tildeortion to $(C_1T_1) : (C_2T_2)$.

MASTER FORMULA CHEATSHEET — NUMERICAL ABILITY & DATA INTERPRETATION

1. Percentages & Multipliers:

$$\text{Percentage Change} = \frac{\text{Final} - \text{Initial}}{\text{Initial}} \times 100\%, \qquad \text{Net Variation} = \left[a + b + \frac{ab}{100}\right]\%$$

$$\text{Reverse Base:} \quad V_{\text{original}} = \frac{V_{\text{final}}}{1 \pm \frac{x}{100}}$$

2. Ratios, Partnerships & Alligation:

$$\text{Share} = \text{Total} \times \left(\frac{\text{Individual Ratio Part}}{\text{Sum of All Parts}}\right), \qquad \text{Profit Ratio} = (C_A T_A) : (C_B T_B)$$

$$\text{Alligation Rule:} \quad \left(\frac{Q_{\text{Cheaper}}}{Q_{\text{Dearer}}}\right) = \frac{D - M}{M - C}$$

3. Averages & Central Tendencies:

$$\text{Sum} = \text{Average} \times N, \qquad A_{\text{weighted}} = \frac{n_1 A_1 + n_2 A_2}{n_1 + n_2}$$

$$\text{New Average} = \text{Old Average} + \frac{V_{\text{new}} - V_{\text{old}}}{N}$$

4. Pie Chart Conversions & Statistics:

$$100\% = 360^\circ \implies 1\% = 3.6^\circ, \qquad \text{Sector Value} = \left(\frac{\text{Angle}}{360}^\circ\right) \times \text{Total Corpus}$$

$$\text{Range} = X_{\max} - X_{\min}, \qquad \text{Empirical Mode} \approx 3 \times \text{Median} - 2 \times \text{Mean}$$

THE SEVEN FATAL CUET TRAPS IN NUMERICAL ABILITY & DI

1. **The Percentage Base Inversion:** Dividing the increase by the final increased value instead of the original initial value, generating an artificially suppressed percentage.
2. **The Successive Symmetry Fallacy:** Assuming that $+20\%$ followed by -20% results in 0% change. The net outcome is always a 4% reduction!
3. **Unsorted Median Blindness:** Picking the middle observation of a raw data array without first arranging the numbers in ascending order.
4. **The Double Pie Chart Distortion:** Comparing interor angles across two different pie charts directly without adjusting for the overall baseline totals of the respective charts.
5. **Averaging the Averages Error:** Calculating the simple average of subgroup means without weighting by the respective population sizes ($n_1 \neq n_2$).
6. **The Metric Scale Blindspot:** Misreading units specified in the chart frame (e.g., treating ₹ in Millions as ₹ in Lakhs, or metric tonnes as quintals).
7. **Subtractive Segment Failure in Stacked Bars:** Reading the total height of a stacked bar instead of subtracting the lower boundary to isolate an internal component's true magnitude.

Practice Questions — Part 1: Percentages & Ratios (Q1 to Q5)

Q1. A retailer increases the marked price of a leather jacket by 25% above its cost price, and subsequently offers a festive discount of 20% on the marked price. The retailer's net profit or loss percentage is:

(A) No profit, no loss (0%) (B) 5% profit
(C) 4% loss (D) 2% profit

Sol. Q1. Let cost price $C = 100$.

- Marked price after 25% hike: $M = 100 \times 1.25 = 125$.
- Discount of 20% on marked price: $D = 125 \times 0.20 = 25$.
- Final Selling Price: $S = 125 - 25 = 100$.

Since Selling Price equals Cost Price ($100 = 100$), the transaction results in **no profit, no loss (0%)** (Option A). Using the successive change formula: $a + b + \frac{ab}{100} = +25 - 20 - \frac{25\times 20}{100} = 5 - 5 = 0\%$. **Answer: (A)**

Q2. In an entrance examination, 65% of candidates passed in Mathematics and 75% passed in English. If 55% of the candidates passed in both subjects and 60 candidates failed in both subjects, the total number of candidates who appeared is:

(A) 300 (B) 350
(C) 400 (D) 450

Sol. Q2. Using the principle of inclusion-exclusion for sets:

$$P(M \cup E) = P(M) + P(E) - P(M \cap E) = 65\% + 75\% - 55\% = 85\%.$$

The proportion of candidates failing in both subjects is $100\% - 85\% = 15\%$. Given that 15% corresponds to 60 candidates:

$$\text{Total Candidates} = \frac{60}{0.15} = 400.$$

Answer: (C)

Q3. A sum of ₹1,440 is divided among three partners P, Q, and R in the ratio $\frac{1}{2} : \frac{1}{3} : \frac{1}{6}$. The share of partner Q is:

(A) ₹360 (B) ₹480
(C) ₹600 (D) ₹720

Sol. Q3. Convert fractional ratio to whole numbers by multiplying through by $\text{LCM}(2, 3, 6) = 6$:

$$P : Q : R = \left(\frac{1}{2} \times 6\right) : \left(\frac{1}{3} \times 6\right) : \left(\frac{1}{6} \times 6\right) = 3 : 2 : 1.$$

Aggregate sum of parts $= 3 + 2 + 1 = 6$. Value of one part $= 1$, $\frac{440}{6} = ₹240$. Partner Q's share $= 2 \times 240 = ₹480$. (Check: P gets $3 \times 240 = 720$, R gets $1 \times 240 = 240$; $720 + 480 + 240 = 1,440$ ✓). **Answer: (B)**

Q4. Two merchants A and B enter into a commercial partnership. Merchant A invests ₹50,000 for 8 months, while Merchant B invests ₹60,000 for 10 months. If the enterprise earns a total annual profit of ₹65,000, Merchant B's share of profit is:

(A) ₹26,000 (B) ₹32,500
(C) ₹39,000 (D) ₹42,000

Sol. Q4. Profit shares are proportional to capital-time product:

$$\text{Profit Ratio} = (C_A \times T_A) : (C_B \times T_B) = (50,000 \times 8) : (60,000 \times 10) = 4,00,000 : 6,00,000 = 2 : 3.$$

Sum of parts $= 2 + 3 = 5$. Unitary share $= 65,\frac{000}{5} = ₹13,000$. Merchant B's share $= 3 \times 13,000 = ₹39,000$. (Merchant A receives $2 \times 13,000 = ₹26,000$; $26,000 + 39,000 = 65,000$ ✓). **Answer: (C)**

Q5. A container holds 80 litres of a mixture of milk and water in the ratio 7 : 1. How many litres of water must be added to the container to make the ratio of milk to water 4 : 1?

(A) 5 litres (B) 7.5 litres
(C) 10 litres (D) 12.5 litres

Sol. Q5. Initial mixture = 80 litres in ratio 7 : 1:

- Total parts $= 7 + 1 = 8$. One part $= \frac{80}{8} = 10$ litres.
- Milk $= 7 \times 10 = 70$ litres; Water $= 1 \times 10 = 10$ litres.

Let x litres of water be added. The volume of milk remains constant at 70 litres:

$$\frac{70}{10 + x} = \frac{4}{1} \implies 4(10 + x) = 70 \implies 40 + 4x = 70 \implies 4x = 30 \implies x = 7.5 \text{ litres.}$$

Answer: (B)

Practice Questions — Part 2: Averages & Statistics (Q6 to Q10)

Q6. The average score of 24 students in a physics quiz was 65. Upon scrutiny, it was discovered that a score of 82 was erroneously recorded as 58. The correct average score of the class is:

(A) 64.0 (B) 65.5
(C) 66.0 (D) 67.5

Sol. Q6. Discrepancy in recorded score:

$$\Delta = \text{Correct Score} - \text{Incorrect Score} = 82 - 58 = +24 \text{ marks.}$$

The aggregate sum of the class must increase by 24 marks. Distribution over 24 students:

$$\Delta A = \frac{+24}{24} = +1.0 \text{ mark.}$$

$$\text{Corrected Average} = 65.0 + 1.0 = 66.0.$$

Answer: (C)

Q7. A batsman has an average of 42 runs in 16 completed innings. In his 17th innings, he scores 90 runs. His new batting average is:

(A) 43.5 runs (B) 44.8 runs
(C) 45.0 runs (D) 46.2 runs

Sol. Q7. Calculate aggregate totals:

- Total runs in 16 innings: $16 \times 42 = 672$ runs.
- Total runs in 17 innings: $672 + 90 = 762$ runs.
- New average $= \frac{762}{17} = 44.82 \approx 44.8$ runs.

Alternative deviation method: The 90 runs exceed the old average by $90 - 42 = 48$ runs. Spread over 17 innings: $\frac{48}{17} \approx 2.82$. New average $= 42 + 2.82 = 44.82$ runs. **Answer: (B)**

Q8. The average age of a husband, wife, and their child 4 years ago was 26 years. The average age of the wife and child 3 years ago was 21 years. The present age of the husband is:

(A) 38 years (B) 40 years
(C) 42 years (D) 44 years

Sol. Q8.

1. Four years ago: Total age of (H + W + C) $= 3 \times 26 = 78$ years.

Present total age of (H + W + C) $= 78 + 3(4) = 78 + 12 = 90$ years.

2. Three years ago: Total age of (W + C) $= 2 \times 21 = 42$ years. Present total age of (W + C) $= 42 + 2(3) = 42 + 6 = 48$ years.

3. Present age of Husband $= 90 - 48 = 42$ years.

Answer: (C)

Q9. The median of the following dataset: 18, 12, 25, 9, 31, 14, 22, 17 is:

(A) 17.0 (B) 17.5
(C) 18.0 (D) 19.5

Sol. Q9. Step 1: Arrange observations in ascending order:

$$9, 12, 14, 17, 18, 22, 25, 31$$

Sample size $N = 8$ (even count). The median is the arithmetic mean of the two middle elements (positions 4 and 5):

- 4th element = 17
- 5th element = 18

$$\text{Median} = \frac{17 + 18}{2} = \frac{35}{2} = 17.5.$$

Answer: (B)

Q10. In a moderately skewed statistical distribution, the mean is 36.0 and the median is 34.0. Using the empirical relationship, the estimated mode of the distribution is:

(A) 30.0 (B) 32.0
(C) 35.0 (D) 38.0

Sol. Q10. Using the empirical Pearsonian relationship:

$$\text{Mode} \approx 3 \times \text{Median} - 2 \times \text{Mean}$$

Substitute the given parameters:

$$\text{Mode} \approx 3(34.0) - 2(36.0) = 102.0 - 72.0 = 30.0.$$

Answer: (A)

Practice Questions — Part 3: Data Interpretation Set (Q11 to Q15)

REFERENCE DATA TABLE FOR Q11 TO Q15

The table below depicts the annual production (in thousand metric tonnes) of five agro-commodities (Wheat, Rice, Maize, Pulses, Barley) across four agrarian states (Punjab, Haryana, UP, MP) in the financial year 2025–26:

State	Wheat	Rice	Maize	Pulses	Total (Th. MT)
Punjab	160	120	30	10	320
Haryana	120	80	40	20	260
UP	240	180	60	40	520
MP	180	60	70	90	400
Total	700	440	200	160	1,500

Q11. What percentage of the total agro-commodity production across all four states is accounted for by Wheat?

(A) 42.5% (B) 44.0%
(C) 46.67% (D) 48.0%

Sol. Q11. Total Wheat production = 700 thousand metric tonnes. Total aggregate agro-commodity production = 1,500 thousand metric tonnes.

$$\text{Percentage} = \left(\frac{700}{1,500}\right) \times 100\% = \frac{7}{15} \times 100\% = 46.67\%.$$

Answer: (C)

Q12. The ratio of the total production of Pulses to the total production of Rice across all four states combined is:

(A) 3 : 8 (B) 4 : 11
(C) 2 : 5 (D) 5 : 12

Sol. Q12. Total Pulses production = 160 thousand MT. Total Rice production = 440 thousand MT.

$$\text{Ratio} = 160 : 440 = 16 : 44 = 4 : 11.$$

Answer: (B)

Q13. Which state records the highest tildeortion of its own total production in the form of Maize?

(A) Punjab (B) Haryana
(C) UP (D) MP

Sol. Q13. Calculate Maize tildeortion for each state:

- Punjab: $\frac{30}{320} = \frac{3}{32} \approx 9.38\%$.
- Haryana: $\frac{40}{260} = \frac{4}{26} = \frac{2}{13} \approx 15.38\%$.
- UP: $\frac{60}{520} = \frac{6}{52} = \frac{3}{26} \approx 11.54\%$.
- MP: $\frac{70}{400} = \frac{7}{40} = 17.50\%$.

MP demonstrates the highest tildeortion at 17.50%. **Answer: (D)**

Q14. By what percentage does the total Wheat production in UP exceed the total Wheat production in Punjab?

(A) 40.0% (B) 45.0%
(C) 50.0% (D) 66.67%

Sol. Q14. Wheat in UP = 240 thousand MT; Wheat in Punjab = 160 thousand MT. Absolute excess = $240 - 160 = 80$ thousand MT. Base value is Punjab = 160 thousand MT:

$$\%\ \text{Excess} = \left(\frac{80}{160}\right) \times 100\% = \frac{1}{2} \times 100\% = 50.0\%.$$

Answer: (C)

Q15. If the production data for MP is represented in a standalone pie chart, the central angle subtended by the interor representing Pulses is:

(A) 72.0° (B) 81.0°
(C) 90.0° (D) 96.0°

Sol. Q15. Total production of MP = 400 thousand MT. Pulses production in MP = 90 thousand MT. Central angle calculation:

$$\text{Angle} = \left(\frac{\text{Component Value}}{\text{Total Value}}\right) \times 360^\circ = \left(\frac{90}{400}\right) \times 360^\circ = \frac{9}{40} \times 360^\circ = 9 \times 9^\circ = 81.0^\circ.$$

Answer: (B)